

Problem Statement:

AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs

Description:

- **Background** The government actively promotes the economic empowerment of marginalized communities by providing concessional credit for income-generating activities. Under various schemes, beneficiaries are required to contribute a small margin money fraction—typically 10% of the total project cost—while the State Channelizing agencies (SCAs) Channelizing agencies(CAs) provide the remaining 90% as a concessional loan. For example, if an entrepreneur wishes to establish a ₹10,00,00 enterprise, they must possess ₹1,00,00 as their 10% contribution, making them eligible for a ₹9,00,00 loan.

These loans are categorized into specific tiers:

Micro Finance Scheme: For small units with a project cost up to ₹1.40 lakh. The funding agency provides up to 90% (maximum ₹1.25 lakh) at a concessional interest rate of 6.5% per annum for the beneficiary, to be repaid over 3 years (including a 3-month moratorium).

Term Loan Scheme: For larger projects costing between ₹1.40 lakh and ₹50.00 lakh. The agency provides up to 90% (maximum ₹45 lakh) at an interest rate of 8% per annum, to be repaid over 7 years (including a 6-month moratorium).

However, despite the availability of capital, many first-time rural entrepreneurs face a high rate of business stagnation. This is due to a profound lack of formal market research tailored to their specific geographical reality, compounded by poor financial literacy regarding loan structuring, margin requirements, and repayment schedules.

- **Challenge**

Beneficiaries often select business activities based on anecdotal success rather than data-driven market demand, and they struggle to calculate exactly how much capital they need or which scheme they qualify for. A prospective entrepreneur in a specific Gram Panchayat (village) lacks the analytical tools to determine local market saturation, optimal pricing, localized threats, and their precise financial eligibility.

There is a critical need for an intelligent tool that democratizes institutional-grade business consulting. The challenge is to build a hyper-local AI Assistant accompanied by a Smart Scheme Calculator that guides the user through a comprehensive business feasibility study and financial structuring plan before they apply for funding.

- **Expected Solution**

Participants are required to develop an NLP-powered, multilingual AI Business Advisory Assistant tailored for rural and semi-urban geographies. The system should take basic inputs from the user:

Geographic Location (Village/Block/District), Available Margin Capital (e.g., ₹1,00,000), and the

Proposed Business Category (e.g., Dairy, Retail, Textiles etc).

The application must feature two core modules:

Module 1: Hyper-Local Business Feasibility Report

The AI must dynamically generate a localized strategy encompassing:

1. Market Reach: Estimating the immediate consumer base within a 5 to 10 km radius of the village/block and identifying primary distribution channels.
2. Opportunity Analysis: Highlighting unserved or underserved niches within the chosen business sector in that specific local economy.
3. General Business Analysis (SWOT): A foundational breakdown of Strengths, Weaknesses, Opportunities, and Threats tailored to the specific micro-enterprise budget.
4. Threats Identification: Pinpointing local risks such as supply chain bottlenecks, seasonal demand fluctuations, or dependency on single buyers.
5. Competitor Mapping: Using localized demographic and economic data to estimate the density of existing similar businesses in the block.
6. Product Market Value: Suggesting optimal pricing strategies and predicting the local market value of the goods/services based on regional purchasing power.

Module 2:

Smart Financial Calculator & Scheme Router An integrated financial engine that automatically processes the user's 'Available Margin Capital' to output a clear financial roadmap: 7.

Financial Structuring:

Automatically calculates the total feasible Project Cost (Available Margin / 10%) and the Maximum Loan Amount (90% of Project Cost).

Example: If the user inputs 1,00,00 as available capital, the tool establishes a ₹10,00,00 Project Cost and a approx. 9,00,00 loan eligibility.

Scheme Auto-Selection:

Routes the user to the correct scheme based on the calculated Project Cost.

Logic A: If Project Cost less than 1.40 Lakh -> Selects Micro Finance Scheme (6.5% interest, 3-year tenure, 3-month moratorium).

Logic B: If Project Cost between 1.40 Lakh and 50.00 Lakh -> Selects Term Loan Scheme (8% interest, 7-year tenure, 6-month moratorium).

EMI & Moratorium Generator: Outlines the expected quarterly repayment schedule, operational costs, and working capital requirements, factoring in the specific moratorium periods.

- Impact Goals

Reduce the failure rate of newly funded micro-enterprises by ensuring beneficiaries choose viable, locally relevant business models based on data.

Eliminate financial confusion by clearly mapping a beneficiary's available cash (10%) to their maximum borrowing capacity (90%) and exact repayment obligations.

Empower marginalized youth with enterprise intelligence, fostering a culture of data-backed, financially sound entrepreneurship at the grassroots level.